Rmk[section]

Josh's MAT223 Notes

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Abstract

Theorems and definitions from the first half of MAT223H1Y in Summer 2025.

0 Interesting Facts

Assignment Questions Facts

Assignment 8

Thm 0.1. (Fundamental Theorem of Algebra, A8Q1)
Let $f(x)$ be a polynomial of degree n with coefficients in $\mathbb{R}$. Then we can write

$$f(x) = (x - a_1)^{m_1} \dots (x - a_k)^{m_k} \tag{F}$$

where $a_1, \dots, a_k$ are distinct complex numbers and $m_i \geq 1$ are integers satisfying $m_1 + \dots + m_k = n$

Lem 0.1. χ_A is a polynomial of degree n.

Prop 0.1. If A and B are similar matrices, then they have the same eigenvalues.

Lem 0.2. If A is diagonalizable, then A^n is diagonalizable.

Assignment 9

Lem 0.3. Non-zero orthogonal vectors are linearly independent.

Prop 0.2. O is an orthogonal matrix $\iff$ $||Ov|| = ||v||$ for any vector $v \in \mathbb{R}^n$.

Prop 0.3. If n is odd, then any $n \times n$ matrix with real entries has at least one real eigenvalue.

1 Linear Equations and Matrices

1.1 Linear equations and systems

Def. [Linear equation] A **linear equation** in variables $x_1, \ldots, x_n$ is an equation of the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

where $a_1, \ldots, a_n, b \in \mathbb{R}$ are fixed real numbers (i.e., do not depend on $x_1, \ldots, x_n$).

Ex. Linear: $x + 2y - 3z = 5$. Not linear: $x^2 + y + z = 1$, $\sin(x) + y = 3$, $xy + z = 0$.

Def. [System of linear equations] A **system of linear equations** is a collection of linear equations in the same variables. A tuple $(s_1, \ldots, s_n) \in \mathbb{R}^n$ is a **solution** to the system if it satisfies every equation in the system.

Ex.

$$\begin{cases} 7x + y + 5z = 27 \\ 4x + 3y + 5z = 21 \\ 6x + y + 2z = 9 \end{cases}$$

The point $(0, -3, 6)$ is a solution since $7 \cdot 0 + 1 \cdot (-3) + 5 \cdot 6 = 27$, $4 \cdot 0 + 3 \cdot (-3) + 5 \cdot 6 = 21$, and $6 \cdot 0 + 1 \cdot (-3) + 2 \cdot 6 = 9$.

1.2 Matrix representation

Def. [$n \times m$ matrix and entries] A **matrix** is a rectangular array of numbers. If it has n rows and m columns, it is an $n \times m$ matrix; its entries are denoted a_{ij} (ith row, jth column).

Def. [Coefficient and augmented matrices] For a system

$$\begin{cases} a_{11}x_1 + \cdots + a_{1m}x_m = b_1 \\ \vdots \\ a_{n1}x_1 + \cdots + a_{nm}x_m = b_n \end{cases}$$

the **coefficient matrix** is $C = (a_{ij}) \in \mathbb{R}^{n \times m}$ and the **augmented matrix** is

$$A = \left[\, C \mid \vec{b} \,\right] = \begin{bmatrix} a_{11} & \cdots & a_{1m} & b_1 \\ \vdots & \ddots & \vdots & \vdots \\ a_{n1} & \cdots & a_{nm} & b_n \end{bmatrix}.$$

*** 1.** There is a one-to-one correspondence between $n \times (m+1)$ augmented matrices (with m variables) and systems of linear equations.

1.3 Solution-preserving operations

Def. [Equivalent systems] Two systems of linear equations are **equivalent** if they have the same solution set.

Prop 1.1. [Solution-preserving equation operations] The following operations produce an equivalent system:

(E1) Interchange two equations in the system.

(E2) Replace an equation by a nonzero scalar multiple of itself.

(E3) Replace an equation by itself plus a nonzero scalar multiple of another equation.

Proof. Each operation is reversible and does not change the set of simultaneous solutions: (E1) just reorders equations; (E2) scales an identity by a nonzero constant; (E3) adds a consequence of the existing equations. Hence the solution set is preserved. □

1.4 Elementary row operations and row equivalence

Def. [Elementary row operations (EROs)] On a matrix we may perform:

(ER1) interchange two rows,

(ER2) multiply a row by a nonzero scalar,

(ER3) replace a row by itself plus a scalar multiple of another row.

Def. [Row equivalence] Two matrices are **row equivalent** if one can be obtained from the other by a sequence of EROs.

Thm 1.1. If two augmented matrices are row equivalent, then the systems of linear equations they represent are equivalent.

Proof. Each ERO corresponds exactly to the solution-preserving operations (E1)–(E3) on the underlying system; therefore the solution set is unchanged after any sequence of EROs. □

Ex. [System ↔ matrices] For

$$\begin{cases} 7x + y + 5z = 27 \\ 4x + 3y + 5z = 21 \\ 6x + y + 2z = 9 \end{cases} \qquad \Longleftrightarrow \qquad \left[\begin{array}{ccc|c} 7 & 1 & 5 & 27 \\ 4 & 3 & 5 & 21 \\ 6 & 1 & 2 & 9 \end{array}\right],$$

EROs on the augmented matrix implement the legal operations on the system and therefore preserve the solution set.

2 Solving Linear Systems & Geometric Implications

2.1 Echelon shape and back–substitution

Prop 2.1. A system of linear equations whose augmented matrix has a "stair–case" (echelon) shape can be solved easily by **back–substitution**. Equations corresponding to lower pivots determine the last variables first, then successively determine the earlier ones.

Def. [Pivot] In a given row of a matrix, the **pivot** is the leftmost nonzero entry of that row.

2.2 Row echelon forms

Def. [Row echelon form (REF)] A matrix is in **row echelon form** if

(i) all zero rows are at the bottom, and

(ii) each pivot lies strictly to the right of the pivot in the row above it.

Def. [Reduced row echelon form (RREF)] A matrix is in **reduced row echelon form** if, in addition to being in REF,

(i) every pivot equals 1, and

(ii) each pivot is the only nonzero entry in its column.

*** 2.** Any system whose matrix is in (reduced) row echelon form can be solved by back–substitution. Thus we should *first* put its augmented matrix into REF/RREF.

2.3 Gauss–Jordan elimination

Def. [Row equivalence & EROs] Matrices A and B are **row equivalent** if one can be obtained from the other by a sequence of elementary row operations: (ER1) swap two rows; (ER2) multiply a row by a nonzero scalar; (ER3) replace a row by itself plus a scalar multiple of another row.

Thm 2.1 (Gauss–Jordan)**.** Every matrix A is row equivalent to a matrix X in (reduced) row echelon form. Moreover, if X is in RREF, then X is *unique*.

Proof sketch (algorithmic). Starting from A: (1) move all zero rows to the bottom (ER1); (2) reorder nonzero rows so pivots step to the right as we go down (ER1); (3) use ER3 to clear entries *below* each pivot (REF reached); (4) scale rows to make pivots 1 (ER2); (5) use ER3 to clear all *nonpivot* entries in pivot columns (RREF reached). Uniqueness follows because each of these requirements determines the same RREF regardless of the sequence of EROs. □

Def. [RREF notation] For a matrix A, we denote its reduced row echelon form by $\mathrm{rref}(A)$.

2.4 Basic/free variables and pivot columns

Let $X = \mathrm{rref}(A)$ for the augmented matrix of a system with unknowns $x_1, \ldots, x_m$.

Def. (1) Column i of the coefficient matrix is a **pivot column** if the ith column of X contains a pivot.

(2) The variable x_i is **basic** if column i is a pivot column.

(3) The variable x_i is **free** if column i is not a pivot column.

*** 3.** A system has infinitely many solutions iff it has at least one free variable. Solutions are parametrized by the free variables.

2.5 Consistency and number of solutions

Def. A system is **consistent** if it has at least one solution; **inconsistent** if it has none.

Thm 2.2 (Rouché–Capelli (pivot test))**.** Let $A = [\, C \mid \vec{b} \,]$ be the augmented matrix and C its coefficient matrix.

(1) The system is **inconsistent** $\iff$ the last column of $\operatorname{rref}(A)$ contains a pivot.

(2) The system has **exactly one** solution $\iff$ the last column of $\operatorname{rref}(A)$ has no pivot and *every* column of $\operatorname{rref}(C)$ is a pivot column (no free variables).

(3) The system has **infinitely many** solutions $\iff$ the last column of $\operatorname{rref}(A)$ has no pivot and at least one column of $\operatorname{rref}(C)$ is *not* a pivot column (some free variable).

Cor 2.1. Every system of linear equations has either no solution, exactly one solution, or infinitely many solutions.

2.6 Geometric picture (intuition)

In two variables, solutions correspond to intersections of lines in $\mathbb{R}^2$: coincident lines $\Rightarrow$ infinitely many; single intersection $\Rightarrow$ one solution; parallel distinct lines $\Rightarrow$ none. In three variables, solutions correspond to intersections of planes in $\mathbb{R}^3$: no common point (none); intersecting in a line (infinitely many); intersecting in a single point (one).

3 Vectors in Euclidean Space

3.1 What is a vector? Standard coordinates

Def. [Vector] A **vector** is a mathematical object (often drawn as an arrow) with a *length* and a *direction*. We may view a vector as the *displacement* from the tail of the arrow to its head.

Def. [Standard coordinates in $\mathbb{R}^2$ and $\mathbb{R}^3$] If a vector in $\mathbb{R}^2$ represents displacement by v_1 in the x-direction and by v_2 in the y-direction, its **standard coordinate representation** is

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}.$$

If a vector in $\mathbb{R}^3$ represents displacement by w_1 in x, w_2 in y, and w_3 in z, its standard coordinates are

$$\vec{w} = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix}.$$

*** 4.** Vectors need not start at the origin; two arrows represent the *same vector* if they have the same length and direction. There is a natural 1–1 correspondence between vectors in $\mathbb{R}^2$ (resp. $\mathbb{R}^3$) and points in the plane (resp. space).

3.2 Vector addition and scalar multiplication

Def. [Operations in $\mathbb{R}^2$] For $\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$, $\vec{w} = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \in \mathbb{R}^2$ and $c \in \mathbb{R}$,

$$\vec{v} + \vec{w} = \begin{pmatrix} v_1 + w_1 \\ v_2 + w_2 \end{pmatrix} \qquad \text{and} \qquad c\,\vec{v} = \begin{pmatrix} cv_1 \\ cv_2 \end{pmatrix}.$$

Def. [Operations in $\mathbb{R}^3$] For $\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$, $\vec{w} = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} \in \mathbb{R}^3$ and $c \in \mathbb{R}$,

$$\vec{v} + \vec{w} = \begin{pmatrix} v_1 + w_1 \\ v_2 + w_2 \\ v_3 + w_3 \end{pmatrix}, \qquad c\,\vec{v} = \begin{pmatrix} cv_1 \\ cv_2 \\ cv_3 \end{pmatrix}.$$

*** 5.** [Geometric meaning] Scalar multiplication $c\vec{v}$ stretches or reverses $\vec{v}$ by factor c. Vector addition $\vec{v} + \vec{w}$ is obtained by the parallelogram (or tip-to-tail) rule.

3.3 Higher-dimensional vectors

Def. [Vectors in $\mathbb{R}^n$ and standard coordinates] Fix $n \in \mathbb{N}$, $n \geq 3$. An n**-dimensional**

vector is an ordered list of n real numbers,

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = (v_1, \ldots, v_n).$$

We call $v_1, \ldots, v_n$ the *standard coordinates* of $\vec{v}$. The set of all n-dimensional vectors is $\mathbb{R}^n$.

Def. [Operations in $\mathbb{R}^n$] For $\vec{v} = (v_1, \ldots, v_n)^T$, $\vec{w} = (w_1, \ldots, w_n)^T \in \mathbb{R}^n$ and $c \in \mathbb{R}$,

$$\vec{v} + \vec{w} = (v_1 + w_1, \ldots, v_n + w_n)^T, \qquad c\,\vec{v} = (cv_1, \ldots, cv_n)^T.$$

3.4 Vector equations and linear systems

Given a system of linear equations, we can collect the coefficients of each variable as a *column vector*. For example, with unknowns $x, y, z \in \mathbb{R}$,

$$x \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + y \begin{pmatrix} -1 \\ -3 \\ 1 \end{pmatrix} + z \begin{pmatrix} -1 \\ -1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ -3 \end{pmatrix}.$$

This is the **vector equation** associated with the system.

Prop 3.1. The vector equation

$$x_1\vec{v}_1 + x_2\vec{v}_2 + \cdots + x_n\vec{v}_n = \vec{w}$$

has the same solution set as the linear system represented by the augmented matrix

$$\left[\, \vec{v}_1 \ \vec{v}_2 \ \cdots \ \vec{v}_n \mid \vec{w} \,\right].$$

Proof. Write $\vec{v}_j = (a_{1j}, \ldots, a_{mj})^T$ and $\vec{w} = (w_1, \ldots, w_m)^T$. Equality of vectors is componentwise, so the vector equation is equivalent to the m scalar equations $\sum_{j=1}^{n} a_{ij}x_j = w_i$ ($i = 1, \ldots, m$), which is exactly the system encoded by the augmented matrix. $\square$

4 Linear Combinations & Systems of Linear Equations

4.1 Linear combinations and span

Def. [Linear combination] Given vectors $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n$, a **linear combination** of them is any vector of the form

$$a_1\vec{v}_1 + a_2\vec{v}_2 + \cdots + a_n\vec{v}_n$$

for some $a_1, \ldots, a_n \in \mathbb{R}$. We call $a_1, \ldots, a_n$ the *coefficients*.

Def. [Span] Let $\vec{v}_1, \ldots, \vec{v}_n \in \mathbb{R}^m$. Then

$$\mathrm{Span}\{\vec{v}_1, \ldots, \vec{v}_n\} = \{a_1\vec{v}_1 + \cdots + a_n\vec{v}_n \mid a_1, \ldots, a_n \in \mathbb{R}\}.$$

In words: $\mathrm{Span}\{\vec{v}_1, \ldots, \vec{v}_n\}$ is the set of *all* linear combinations of $\vec{v}_1, \ldots, \vec{v}_n$.

Ex. (1) In $\mathbb{R}^2$, if $\vec{v} \neq \vec{0}$, then $\mathrm{Span}\{\vec{v}\} = \{t\vec{v} \mid t \in \mathbb{R}\}$ is a line through the origin.
(2) In $\mathbb{R}^3$, if $\vec{v}_1, \vec{v}_2$ are not scalar multiples, then

$$\mathrm{Span}\{\vec{v}_1, \vec{v}_2\} = \{t\vec{v}_1 + s\vec{v}_2 \mid s, t \in \mathbb{R}\}$$

is a plane through the origin.
(3) In solving a linear system: if $A = [\vec{A}_1 \ \cdots \ \vec{A}_m]$ and $\vec{b} \in \mathbb{R}^n$, then a solution exists only if $\vec{b} \in \mathrm{Span}\{\vec{A}_1, \ldots, \vec{A}_m\}$.

Do n vectors always span an n-dimensional object? No. For instance in $\mathbb{R}^3$, take $\vec{v}_2 = 2\vec{v}_1 + \vec{v}_3$. Then $\mathrm{Span}\{\vec{v}_1, \vec{v}_2, \vec{v}_3\} = \mathrm{Span}\{\vec{v}_1, \vec{v}_3\}$, which is at most a plane (not all of $\mathbb{R}^3$).

4.2 Linear (in)dependence

Def. [Geometric definition] Given a list $\vec{v}_1, \ldots, \vec{v}_n$, we say they are **linearly dependent** if, for at least one i,

$$\vec{v}_i \in \mathrm{Span}\{\vec{v}_1, \ldots, \vec{v}_{i-1}, \vec{v}_{i+1}, \ldots, \vec{v}_n\}.$$

Otherwise, the vectors are **linearly independent**.

Def. [Algebraic definition] Vectors $\vec{v}_1, \ldots, \vec{v}_n$ are **linearly dependent** if the vector equation

$$a_1\vec{v}_1 + \cdots + a_n\vec{v}_n = \vec{0}$$

has a nontrivial solution (some $a_i \neq 0$). Otherwise they are **linearly independent**.

Prop 4.1. The geometric and algebraic definitions of linear (in)dependence are equivalent.

Proof. If $\vec{v}_i = \sum_{j \neq i} a_j\vec{v}_j$ with some coefficient nonzero, then $(-1)\vec{v}_i + \sum_{j \neq i} a_j\vec{v}_j = \vec{0}$ is a nontrivial relation. Conversely, from a nontrivial relation $\sum a_i\vec{v}_i = \vec{0}$ pick an index with $a_i \neq 0$ and solve for $\vec{v}_i$ to express it as a combination of the others. $\square$

4.3 Testing (in)dependence via equations and matrices

Prop 4.2. The vector equation

$$x_1\vec{v}_1 + x_2\vec{v}_2 + \cdots + x_n\vec{v}_n = \vec{w}$$

has the same solution set as the linear system represented by the augmented matrix with columns $[\ \vec{v}_1\ \vec{v}_2\ \cdots\ \vec{v}_n \mid \vec{w}\ ]$.

Proof. Equate components to obtain n scalar equations $\sum_{j=1}^{n} a_{ij}x_j = w_i$ where $\vec{v}_j = (a_{1j}, \ldots, a_{nj})^T$ and $\vec{w} = (w_1, \ldots, w_n)^T$. This is exactly the system encoded by the augmented matrix; conversely stacking those equations recovers the vector equation. □

Ex. Are $(1,1)^T, (1,0)^T, (2,1)^T$ linearly independent? Form the coefficient matrix with these as columns:

$$C = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \end{pmatrix}.$$

Row-reduction gives rank 2 (one free variable), so the homogeneous system $Cx = \vec{0}$ has infinitely many solutions. Hence the three vectors are linearly *dependent*.

Ex. For $\mathbb{R}^3$-vectors $(1,1,0)^T, (3,1,2)^T, (-1,2,0)^T$, the 3×3 matrix with these columns row-reduces to an upper-triangular matrix with three pivots (equivalently, has nonzero determinant), so the vectors are linearly *independent*.

4.4 Matrix–vector product

Def. [Matrix–vector product] Let A be an $n \times m$ matrix with columns $\vec{A}_1, \ldots, \vec{A}_m$ and let $\hat{x} = (x_1, \ldots, x_m)^T \in \mathbb{R}^m$. The product $A\hat{x} \in \mathbb{R}^n$ is defined by

$$A\hat{x} = x_1\vec{A}_1 + \cdots + x_m\vec{A}_m$$

(equivalently, $(A\hat{x})_i = \sum_{j=1}^{m} a_{ij}x_j$ for $i = 1, \ldots, n$).

*** 6.** Conformability: $A\hat{x}$ is defined only when the length of $\hat{x}$ equals the number of columns of A (i.e. m). Then $A\hat{x} \in \mathbb{R}^n$, where n is the number of rows of A.

Ex. Let

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ -1 & 1 \end{pmatrix}\ (3 \times 2), \quad \vec{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}\ (2 \times 1), \quad \vec{u} = \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}\ (3 \times 1).$$

Then $A\vec{v}$ is well-defined and equals

$$A\vec{v} = 2\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + 1\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ -1 \end{pmatrix}$$

(which also matches the component formula). The product $A\vec{u}$ is *not* defined since the sizes do not match.

4.5 Four equivalent representations of a linear system

Given $A \in \mathbb{R}^{n\times m}$ and $\vec{b} \in \mathbb{R}^n$, the system of linear equations can be written as:

(1) **Algebraic form:**

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1m}x_m = b_1 \\ \qquad \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nm}x_m = b_n \end{cases}$$

(2) **Geometric / vector equation:** $x_1\vec{A}_1 + \cdots + x_m\vec{A}_m = \vec{b}$. A solution exists $\iff$ $\vec{b} \in \mathrm{Span}\{\vec{A}_1, \ldots, \vec{A}_m\}$.

(3) **Augmented matrix:** $[\ A \mid \vec{b}\ ]$. Row-reduction (R–C) reveals existence/number of solutions via pivots.

(4) **Matrix–vector equation:** $A\hat{x} = \vec{b}$.

*** 7.** [Motivation] We can view A as defining a linear map $F : \mathbb{R}^m \to \mathbb{R}^n$ by $F(\hat{x}) = A\hat{x}$.

Ex. The vector equation $a\binom{1}{2} + b\binom{2}{1} = \binom{2}{1}$ is the matrix–vector system

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

5 Vector Subspaces in $\mathbb{R}^n$

5.1 Reminders

Vectors in $\mathbb{R}^n$ add and scale component-wise. For $v = (v_1, \dots, v_n)^T$, $w = (w_1, \dots, w_n)^T$ and $c \in \mathbb{R}$,

$$v + w = (v_1 + w_1, \dots, v_n + w_n)^T, \qquad cv = (cv_1, \dots, cv_n)^T.$$

For $v_1, \dots, v_k \in \mathbb{R}^n$,

$$\operatorname{Span}\{v_1, \dots, v_k\} = \{a_1 v_1 + \cdots + a_k v_k : a_1, \dots, a_k \in \mathbb{R}\}.$$

Example: the x–y plane in $\mathbb{R}^3$ is $\operatorname{Span}\{(1,0,0)^T, (0,1,0)^T\}$.

Prop 5.1. ["Subspace test"] If a subset $V \subseteq \mathbb{R}^n$ equals $\operatorname{Span}\{v_1, \dots, v_k\}$, then V is *closed under scalar multiplication* and *closed under vector addition.*

Def. [Vector subspace of $\mathbb{R}^n$] A **vector subspace** V of $\mathbb{R}^n$ is a non-empty subset of $\mathbb{R}^n$ such that

(i) (Scalar multiplication) If $v \in V$ and $c \in \mathbb{R}$, then $cv \in V$.

(ii) (Vector addition) If $v, w \in V$, then $v + w \in V$.

(Consequently $0 \in V$ since $0 \cdot v = 0$.)

Ex. A plane through the origin in $\mathbb{R}^3$ is a vector subspace: it is $\operatorname{Span}\{v_1, v_2\}$ for some v_1, v_2. By contrast $S = \{(1, y, 0)^T : y \in \mathbb{R}\}$ is not a subspace of $\mathbb{R}^3$ since $2(1,0,0)^T = (2,0,0)^T \notin S$.

Lem 5.1. If S is a set of n linearly independent vectors in $\mathbb{R}^m$, then $n \le m$.

Thm 5.1 (Characterization by span)**.** A subset $V \subseteq \mathbb{R}^m$ is a vector subspace of $\mathbb{R}^m$ if and only if there exist $v_1, \dots, v_n \in \mathbb{R}^m$ such that $V = \operatorname{Span}\{v_1, \dots, v_n\}$.

Proof. ($\Leftarrow$) If $V = \operatorname{Span}\{v_1, \dots, v_n\}$, then $0 = 0v_1 + \cdots + 0v_n \in V$. If $v = \sum a_i v_i \in V$ and $c \in \mathbb{R}$, then $cv = \sum (ca_i) v_i \in V$. If $v = \sum a_i v_i$ and $w = \sum b_i v_i$, then $v + w = \sum (a_i + b_i) v_i \in V$. Hence V is a subspace.

($\Rightarrow$) Suppose V is a subspace. Let $B = (v_1, \dots, v_n)$ be a longest list of linearly independent vectors in V (such a list exists and has $n \le m$ by the lemma). Then $\operatorname{Span} B \subseteq V$. For $w \in V$, if $w = 0$ or $w = v_i$ for some i we are done. Otherwise, since B is maximal, the list $(v_1, \dots, v_n, w)$ is linearly dependent, so there are scalars $a_1, \dots, a_n, b$, not all zero, with $a_1 v_1 + \cdots + a_n v_n + bw = 0$. Because $v_1, \dots, v_n$ are independent, $b \neq 0$, hence $w = -(a_1/b)v_1 - \cdots - (a_n/b)v_n \in \operatorname{Span} B$. Thus $V \subseteq \operatorname{Span} B$ and $V = \operatorname{Span} B$. $\square$

Def. [Spanning / generating set] If V is a subspace and $V = \operatorname{Span}\{v_1, \dots, v_n\}$, then $(v_1, \dots, v_n)$ is a **spanning (generating) set** of V.

Ex. Let

$$V = \Big\{(x - y,\ 2x + y,\ 3y)^T :\ x, y \in \mathbb{R}\Big\}.$$

Then

$$(x - y, 2x + y, 3y)^T = x(1,2,0)^T + y(-1,1,3)^T,$$

so $V = \operatorname{Span}\{(1,2,0)^T, (-1,1,3)^T\}$ and is a subspace of $\mathbb{R}^3$.

5.2 Bases

Def. [Basis] Let V be a vector subspace of $\mathbb{R}^m$. A subset $B = \{v_1, \ldots, v_n\} \subseteq \mathbb{R}^m$ is a **basis** of V if it is a linearly independent spanning set of V; that is, $v_1, \ldots, v_n$ are linearly independent and $\mathrm{Span}\{v_1, \ldots, v_n\} = V$.

Ex. To decide whether a list (v_1, v_2, v_3) is a basis of $\mathbb{R}^3$, form the 3×3 matrix with columns v_i. It is a basis iff the matrix reduces to I_3 (equivalently, all columns are pivot columns), i.e., the list is linearly independent and spans $\mathbb{R}^3$.

Thm 5.2 (Uniqueness of basis size)**.** Let V be a vector subspace of $\mathbb{R}^n$. Then the *size* (length) of a basis of V is unique.

Proof. Suppose $(u_1, \ldots, u_k)$ and $(v_1, \ldots, v_\ell)$ are both bases of V with $k < \ell$. Because $(u_1, \ldots, u_k)$ spans V, each $v_i = \sum_{j=1}^{k} a_{ji} u_j$. Let $A = [a_{ji}]$ $(k \times \ell)$ and consider $Ab = 0$. If (v_i) is linearly independent, the only solution is $b = 0$, so every column of A would be a pivot column. But a $k \times \ell$ matrix cannot have ℓ pivot columns when $k < \ell$. Contradiction. Hence $k = \ell$. $\square$

Def. [Dimension] If V is a vector subspace of $\mathbb{R}^n$, the **dimension** of V, denoted $\dim V$, is the size of (the length of) any basis of V. By convention, if $V = \{0\}$, then $\dim V := 0$.

6 Finding a Basis; Column, Row, and Null Spaces

6.1 Finding a Basis

Recall. Let V be a vector subspace of $\mathbb{R}^n$. A set $B = \{\vec{v}_1, \ldots, \vec{v}_m\}$ is a **basis** of V if:

1. B spans V
2. B is linearly independent

Question: Given a list of vectors that spans V but may not be linearly independent, how can we find a basis? We delete the non-pivot columns from the matrix until the remaining columns are linearly independent and still span V.

Thm 6.1. Let A be an $n \times m$ matrix of the form

$$A = \begin{bmatrix} \vec{v}_1 & \vec{v}_2 & \ldots & \vec{v}_m \end{bmatrix}$$

where each $\vec{v}_i \in \mathbb{R}^n$. Suppose

$$\operatorname{rref}(A) = \begin{bmatrix} \vec{x}_1 & \vec{x}_2 & \ldots & \vec{x}_m \end{bmatrix}.$$

If the kth column of $\operatorname{rref}(A)$ is not a pivot column, then $\vec{v}_k \in \operatorname{Span}(\vec{v}_1, \ldots, \widehat{\vec{v}_k}, \ldots, \vec{v}_m)$ and can be removed without changing the span.

Proof. If $\vec{v}_k$ is not a pivot column, the corresponding variable c_k in the system

$$c_1\vec{v}_1 + \cdots + c_k\vec{v}_k + \cdots + c_m\vec{v}_m = 0$$

is free. Setting $c_k = 1$ and solving for the other c_i expresses $\vec{v}_k$ as a linear combination of the pivot columns. □

Thm 6.2. Let $V = \operatorname{Span}(\vec{v}_1, \ldots, \vec{v}_m)$. If $A = [\vec{v}_1 \ldots \vec{v}_m]$ has k pivot columns, then those pivot columns form a basis for V, and $\dim(V) = k$.

6.2 Three Fundamental Subspaces of a Matrix

Recall. If A is an $m \times n$ matrix and $\vec{x} \in \mathbb{R}^n$, then

$$A\vec{x} = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + \cdots + a_{1n}x_n \\ a_{21}x_1 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n \end{bmatrix}.$$

Def. [Column, Null, and Row spaces] Let $A \in \mathbb{R}^{m \times n}$. Then:

- **Column space:** $\operatorname{col}(A) = \operatorname{Span}\{\text{columns of } A\} \subseteq \mathbb{R}^m$
- **Null space:** $\operatorname{null}(A) = \{\vec{x} \in \mathbb{R}^n : A\vec{x} = 0\} \subseteq \mathbb{R}^n$
- **Row space:** $\operatorname{row}(A) = \operatorname{Span}\{\text{rows of } A\} \subseteq \mathbb{R}^n$

Remark:

$$\text{row}(A) = \text{col}(A^T)$$

i.e., the row space of A is the column space of A^T.

Ex. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}.$$

Then:

$$\text{col}(A) = \text{Span}\left\{ \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \end{bmatrix} \right\} \subseteq \mathbb{R}^2,$$

$$\text{row}(A) = \text{Span}\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \right\} \subseteq \mathbb{R}^3.$$

To find null(A), solve $A\vec{x} = 0$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Row reducing:

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & -3 & -6 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & 0 \end{bmatrix}.$$

From $x_2 + 2x_3 = 0$ we have $x_2 = -2t$, and from $x_1 + 2x_2 + 3x_3 = 0$ we get $x_1 = t$. Thus:

$$\text{null}(A) = \left\{ \begin{bmatrix} t \\ -2t \\ t \end{bmatrix} : t \in \mathbb{R} \right\} = \text{Span}\left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \right\}.$$

7 Rank–Nullity; Solution Sets of Linear Systems

7.1 Column space, null space, rank, nullity

Def. [Column/Null spaces] For an $m \times n$ matrix $A = [\vec{a}_1\, \vec{a}_2\, \dots\, \vec{a}_n]$,

$$\mathrm{col}(A) = \mathrm{Span}\{\vec{a}_1, \dots, \vec{a}_n\} \subseteq \mathbb{R}^m, \qquad \mathrm{null}(A) = \{\vec{x} \in \mathbb{R}^n : A\vec{x} = \vec{0}\} \subseteq \mathbb{R}^n.$$

Def. [Rank/Nullity]

$$\mathrm{rank}(A) = \dim(\mathrm{col}(A)), \qquad \mathrm{nullity}(A) = \dim(\mathrm{null}(A)).$$

Ex. [Quick computation] Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \in \mathbb{R}^{2\times 3}$. Then

$$\mathrm{col}(A) = \mathrm{Span}\left\{\begin{pmatrix} 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 2 \\ 5 \end{pmatrix}, \begin{pmatrix} 3 \\ 6 \end{pmatrix}\right\} \subseteq \mathbb{R}^2, \quad \mathrm{null}(A) = \left\{\begin{bmatrix} -t \\ 2t \\ -t \end{bmatrix} : t \in \mathbb{R}\right\} = \mathrm{Span}\left\{\begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}\right\}.$$

Hence $\mathrm{rank}(A) = 2$ and $\mathrm{nullity}(A) = 1$, which add to $n = 3$ (the number of columns).

7.2 Rank–Nullity Theorem

Thm 7.1 (Rank–Nullity)**.** Let A be an $m \times n$ matrix and let r be the number of pivot columns of A (equivalently, of $\mathrm{rref}(A)$). Then

$$\mathrm{rank}(A) = r \qquad \text{and} \qquad \mathrm{nullity}(A) = n - r.$$

Idea of proof. Write $\mathrm{rref}(A) = [\,\vec{e}_1\ \cdots\ \vec{e}_r\ \mid\ \vec{c}_{r+1}\ \cdots\ \vec{c}_n\,]$ with the first r columns pivot and the rest nonpivot.

(1) The pivot columns of A are linearly independent and span $\mathrm{col}(A)$, so $\mathrm{rank}(A) = r$.

(2) Solving $A\vec{x} = \vec{0}$ (equivalently, $\mathrm{rref}(A)\vec{x} = \vec{0}$), the variables corresponding to nonpivot columns are *free*: $x_{r+1}, \dots, x_n \in \mathbb{R}$. The leading variables are linear combinations

$$x_i = b_{i,r+1}x_{r+1} + \cdots + b_{i,n}x_n, \qquad i = 1, \dots, r,$$

so

$$\vec{x} = x_{r+1}\begin{bmatrix} -b_{1,r+1} \\ \vdots \\ -b_{r,r+1} \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \cdots + x_n\begin{bmatrix} -b_{1,n} \\ \vdots \\ -b_{r,n} \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}.$$

Thus $\mathrm{null}(A)$ is spanned by $n - r$ special solutions, hence $\mathrm{nullity}(A) = n - r$. □

Cor 7.1. For every $m \times n$ matrix A,

$$\mathrm{rank}(A) + \mathrm{nullity}(A) = n.$$

7.3 Homogeneous systems

Def. [Homogeneous system] A linear system is **homogeneous** if its right-hand side is $\vec{0}$, i.e. $A\vec{x} = \vec{0}$.

Thm 7.2. The solution set of any homogeneous linear system is a vector subspace. Moreover, if the coefficient matrix is A, then the solution set equals $\text{null}(A)$.

Proof. Solutions of $A\vec{x} = \vec{0}$ form $\text{null}(A)$, which is a subspace by linearity of A. □

Remark. Knowing $\text{null}(A)$ immediately describes all solutions of the homogeneous system $A\vec{x} = \vec{0}$.

7.4 Inhomogeneous systems and geometry

Def. [Inhomogeneous system] A linear system $A\vec{x} = \vec{b}$ is **inhomogeneous** if $\vec{b} \neq \vec{0}$ (i.e. not homogeneous).

Thm 7.3 (Structure of solutions to $A\vec{x} = \vec{b}$)**.** Assume $A\vec{x} = \vec{b}$ is consistent and let $\vec{p}$ be a particular solution ($A\vec{p} = \vec{b}$). Then

$$\{\text{all solutions}\} = \vec{p} + \text{null}(A) = \{\ \vec{p} + \vec{v}\ :\ \vec{v} \in \text{null}(A)\ \}.$$

Proof. If $A\vec{x} = \vec{b}$ and $A\vec{p} = \vec{b}$, then $A(\vec{x} - \vec{p}) = \vec{0}$, so $\vec{x} - \vec{p} \in \text{null}(A)$ and $\vec{x} \in \vec{p} + \text{null}(A)$. Conversely, if $\vec{v} \in \text{null}(A)$, then $A(\vec{p} + \vec{v}) = A\vec{p} + A\vec{v} = \vec{b} + \vec{0} = \vec{b}$, so $\vec{p} + \vec{v}$ is a solution. □

Remark (geometry). If V is a subspace (e.g., a line or plane through the origin), then sets of the form $\vec{p} + V$ are *translated subspaces* (affine subspaces). In general $\vec{p} + V$ is not a vector space unless $\vec{p} = \vec{0}$. For a consistent inhomogeneous system $A\vec{x} = \vec{b}$, the solution set is the translate $\vec{p} + \text{null}(A)$: a line/plane/etc. parallel to $\text{null}(A)$.

8 Linear Transformations

Motivation

In a finite-dimensional vector space $V = \text{span}\{\vec{v}_1, \ldots, \vec{v}_n\}$, we can think of it geometrically as a *flat surface* (or a line if $\dim(V) = 1$) through the origin.

A **linear transformation** is a function from $\mathbb{R}^n \to \mathbb{R}^m$ that takes a vector subspace of $\mathbb{R}^n$ to a vector subspace of $\mathbb{R}^m$. Geometrically, it takes flat surfaces to flat surfaces.

Def. A function $F : \mathbb{R}^n \to \mathbb{R}^m$ is **linear** if:

1. For any $\vec{x}, \vec{y} \in \mathbb{R}^n$, $F(\vec{x} + \vec{y}) = F(\vec{x}) + F(\vec{y})$ (Additivity)
2. For any $c \in \mathbb{R}$ and $\vec{x} \in \mathbb{R}^n$, $F(c\vec{x}) = cF(\vec{x})$ (Homogeneity)

Example 2

Let $H : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by

$$H\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} \sqrt{x_1} \\ \sqrt{x_2} \end{bmatrix}.$$

Check homogeneity:

$$H\left(c\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \begin{bmatrix} \sqrt{cv_1} \\ \sqrt{cv_2} \end{bmatrix} \neq c\begin{bmatrix} \sqrt{v_1} \\ \sqrt{v_2} \end{bmatrix} \quad \text{(in general)}.$$

Fails condition (2), so H is not linear.

8.1 Matrix Transformations

Recall:

Let A be an $m \times n$ matrix. Then A defines a function from $\mathbb{R}^n$ to $\mathbb{R}^m$ given by

$$A\vec{x} = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} x_1 + \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} x_2 + \cdots + \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} x_n$$

for any $\vec{x} \in \mathbb{R}^n$.

Def. [Matrix Transformation] Let A be an $m \times n$ matrix. The **matrix transformation** T_A of A is the function from $\mathbb{R}^n$ to $\mathbb{R}^m$ given by

$$T_A(\vec{x}) = A\vec{x}, \quad \text{for any } \vec{x} \in \mathbb{R}^n.$$

Prop 8.1. Every matrix transformation is a linear transformation.

Proof. Let $A = \begin{bmatrix} \vec{a}_1 & \ldots & \vec{a}_n \end{bmatrix}$ be an $m \times n$ matrix. We want to show that for any $\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$, $\vec{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n$ and $c \in \mathbb{R}$:

1. $T_A(\vec{x}+\vec{y}) = T_A(\vec{x}) + T_A(\vec{y})$:

$$T_A\left(\begin{bmatrix} x_1+y_1 \\ \vdots \\ x_n+y_n \end{bmatrix}\right) = A\begin{bmatrix} x_1+y_1 \\ \vdots \\ x_n+y_n \end{bmatrix} = A\vec{x} + A\vec{y} = T_A(\vec{x}) + T_A(\vec{y}).$$

2. $T_A(c\vec{x}) = cT_A(\vec{x})$:

$$T_A\left(\begin{bmatrix} cx_1 \\ \vdots \\ cx_n \end{bmatrix}\right) = A\begin{bmatrix} cx_1 \\ \vdots \\ cx_n \end{bmatrix} = c(A\vec{x}) = cT_A(\vec{x}).$$

By the definition of a linear transformation, T_A is linear. $\square$

9 Defining Matrix, Composition of Linear Maps, Rank-Nullity Theorem

9.1 Defining Matrix

Recall:

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

be an $m \times n$ matrix.

A is a map from $\mathbb{R}^n \to \mathbb{R}^m$ given by

$$A\vec{x} = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} x_1 + \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} x_2 + \cdots + \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} x_n$$

for any $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$.

Given a linear map F from a finite-dimensional space to another finite-dimensional space, we want to find a matrix A such that

$$T_A(\vec{x}) = F(\vec{x}).$$

9.1.1 Defining Matrix

Motivation

Let $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ be a basis of $\mathbb{R}^2$. Assume $F(\vec{e}_1) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $F(\vec{e}_2) = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, with $F : \mathbb{R}^2 \to \mathbb{R}^2$ linear.

1. Find $F\left(\begin{bmatrix} 2 \\ 5 \end{bmatrix}\right)$.

$$\begin{aligned} \begin{bmatrix} 2 \\ 5 \end{bmatrix} &= 2\vec{e}_1 + 5\vec{e}_2, \\ F\left(\begin{bmatrix} 2 \\ 5 \end{bmatrix}\right) &= 2F(\vec{e}_1) + 5F(\vec{e}_2) \\ &= 2\begin{bmatrix} 2 \\ 1 \end{bmatrix} + 5\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 19 \\ 22 \end{bmatrix}. \end{aligned}$$

2. Find $F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)$ for any $x, y \in \mathbb{R}$.

$$\begin{aligned} F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) &= xF(\vec{e}_1) + yF(\vec{e}_2) \\ &= x\begin{bmatrix} 2 \\ 1 \end{bmatrix} + y\begin{bmatrix} 3 \\ 4 \end{bmatrix} \\ &= \begin{bmatrix} 2x+3y \\ x+4y \end{bmatrix} \\ &= \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}. \end{aligned}$$

9.2 Standard Basis

Def. [Standard Basis] Given $n \in \mathbb{N}$, the **standard basis** of $\mathbb{R}^n$ is $\{\vec{e}_1, \ldots, \vec{e}_n\}$ where $\vec{e}_i$ is the vector with 1 in the ith coordinate and 0 elsewhere.

Ex.

$$\mathbb{R}^2 : \{\vec{e}_1 = (1,0), \vec{e}_2 = (0,1)\}, \quad \mathbb{R}^4 : \{\vec{e}_1 = (1,0,0,0), \ldots, \vec{e}_4 = (0,0,0,1)\}.$$

Thm 9.1. Every linear transformation is a matrix transformation. In particular, if $F : \mathbb{R}^n \to \mathbb{R}^m$, then $F = T_A$ for some matrix A given by

$$A = \begin{bmatrix} F(\vec{e}_1) & F(\vec{e}_2) & \ldots & F(\vec{e}_n) \end{bmatrix}$$

where $\{\vec{e}_1, \ldots, \vec{e}_n\}$ is the standard basis of $\mathbb{R}^n$.

Proof. Let $\vec{x} \in \mathbb{R}^n$ be arbitrary. Write

$$\vec{x} = x_1\vec{e}_1 + x_2\vec{e}_2 + \cdots + x_n\vec{e}_n.$$

By linearity,

$$F(\vec{x}) = x_1F(\vec{e}_1) + \cdots + x_nF(\vec{e}_n) = \begin{bmatrix} F(\vec{e}_1) & \ldots & F(\vec{e}_n) \end{bmatrix}\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}.$$

Thus $F = T_A$ with A as above. □

9.3 Composition and Matrix Product

Recall. For functions $f : A \to B$ and $g : B \to C$, $(g \circ f)(a) = g(f(a))$.

Let A be $m \times k$ and B be $k \times n$. Then

$$T_A \circ T_B(\vec{x}) = A(B\vec{x}).$$

Def. The **product** AB is the $m \times n$ matrix whose (i,j)th entry is

$$C_{ij} = \sum_{\ell=1}^{k} A_{i\ell}B_{\ell j}.$$

Ex.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 \\ 0 & -1 \\ 3 & 5 \end{bmatrix}, \quad C = AB = \begin{bmatrix} 10 & 15 \\ 22 & 33 \end{bmatrix}.$$

9.4 Rank-Nullity Theorem

Def. For $F : \mathbb{R}^n \to \mathbb{R}^m$ linear, the **kernel** is

$$\ker(F) = \{\vec{x} \in \mathbb{R}^n \mid F(\vec{x}) = \vec{0}\},$$

and the **image** is

$$\mathrm{Im}(F) = \{\vec{y} \in \mathbb{R}^m \mid F(\vec{x}) = \vec{y} \text{ for some } \vec{x} \in \mathbb{R}^n\}.$$

Prop 9.1. [Geometric Rank-Nullity] If $F : \mathbb{R}^n \to \mathbb{R}^m$ is linear, then

$$n = nullity(F) + rank(F).$$

space